

<https://doi.org/10.1038/s41545-026-00616-0>

DFT and Monte Carlo investigation and simulations of PA@Palygorskite nanosorbent in cadmium removal from water



Mohamed E. Mahmoud¹ ✉, Gehan M. Nabil², Mohammed K. Obada³ & Magda E. Abouelanwar¹

Removing cadmium pollutant from wastewater by natural adsorbents represents an eco-friendly and promising strategy. Materials like clay often suffer from low-moderate adsorption capacity, limited stability, and poor reusability, which restrict their practical use. To overcome these challenges, the current study is directed to focus on enhancing the performance of thermally treated Palygorskite (Pal₅₀₀) clay via functionalization with pentetic acid (PA) to generate a novel nanosorbent (PA@Pal₅₀₀). This was extensively characterized to assess its structural and surface properties. Adsorptive removal studies were optimized at pH (1–7), PA@Pal₅₀₀ dosage (10–100 mg), temperature (15–55 °C), contact time (2–120 min), initial Cd(II) concentrations (1.0 and 5.0 mg/L), and ionic strength. Thermodynamic analysis indicated that Cd(II) adsorption by PA@Pal₅₀₀ was correlated to endothermic and spontaneous reactions. Furthermore, PA@Pal₅₀₀ exhibited excellent regeneration capability, confirming its successive reusability. The efficiency of PA@Pal₅₀₀ in the removal of Cd(II) pollutant was validated by real polluted samples, achieving 92.0–95.0% efficiency. Computational results from both DFT and Monte Carlo simulations showed that Cd(II) could bind favorably to Pal₅₀₀ as well as to PA. However, Pal₅₀₀ provided more stable adsorption energies along with a wider variety of active sites on its surface. In contrast, PA tended to coordinate with Cd(II) in a more focused way, mainly through its oxygen-based donor groups, with some additional contribution from nitrogen atoms. Collective findings elucidate that PA@Pal₅₀₀ is a highly effective and sustainable material for cadmium pollutant removal from contaminated water, offering strong potential for real-world environmental remediation applications.

Environmental pollution is listed today as the most issue of global interests challenges and. It occurs when harmful substances are introduced into natural ecosystems and thus negatively affect air, water, soil, and human health^{1–3}. Beyond these direct impacts, pollution also worsens other environmental problems, such as climate change^{4–7}. The driving force for this issue is human activities, including industrial production, urban expansion, traffic congestion, and rapid population growth, all of which are contributing to such air, water, and soil contamination problems⁸. Among these, industrial operations stand out as a major source of pollution via releasing significant and extensive amounts of pollutants, particularly toxic

gases and liquid waste that degrade air and water quality^{9,10}. The released pollutants are commonly categorized as hazardous organic materials and toxic heavy metals like mercury, lead, arsenic, and cadmium^{11–14}.

Cadmium (Cd(II)) is among the most dangerous pollutants, posing serious threats to humans and the environment. It typically enters ecosystems through industrial waste, including runoff from ore mining, galvanizing processes, metal manufacturing, and even wastewater treatment facilities in urban areas¹⁵. High levels of Cd(II) exposure can impose severe health complications such as kidney damage, genetic toxicity, neurological disorders, cancer, and even death¹⁶. The US Environmental Protection

¹Faculty of Sciences, Alexandria University, Alexandria, Egypt. ²Department of Chemistry, College of Science and Humanities in Al-Kharj, Prince Sattam Bin Abdulaziz University, Al-Kharj, Saudi Arabia. ³Egyptian Projects Operation & Maintenance Co. (EPROM), Petroleum complex, Alexandria, Egypt.

✉ e-mail: memahmoud10@yahoo.com

Agency (EPA) has set the maximum allowable concentration of Cd(II) in water at $3.0 \mu\text{g/L}$ ¹⁷.

To tackle Cd(II) contamination in water, a series of treatment techniques has been investigated, including precipitation¹⁸, solvent extraction¹⁹, ion-exchange²⁰, electrochemical processes²¹, membrane filtration²², and biosorption²³. Among these options, the adsorption technique has been listed as a preferred method because of its simplicity, affordability, and high uptake efficacy, making it a practical solution for removing cadmium^{24–26}. Recently, natural clays, especially Palygorskite (commonly known as attapulgite) has attracted considerable attention as promising adsorbents for heavy metal removal owing to their popularity stemming from low cost, wide availability, and excellent adsorption properties²⁷. Palygorskite, also known as fuller's earth, is composed of magnesium-aluminum silicate ($\text{Mg, Al}_2\text{Si}_4\text{O}_{10}(\text{OH})\cdot 4(\text{H}_2\text{O})$). This clay has attracted considerable research interest thanks to its unique structural properties and versatile applications²⁸. Notably, Palygorskite has shown strong potential for removing heavy metals such as cadmium ions, offering an efficient, fast, and cost-effective approach to environmental cleanup processes²⁹. Recent studies have further advanced the use of Palygorskite materials and composites in the removal of toxic pollutants from aquatic systems, demonstrating improved performance and reinforcing its role as a reliable adsorbent^{30–35}.

This study is focused on the batch adsorption of cadmium ions by a novel material constructed via chemical modification of thermally treated Palygorskite at 500°C (Pal_{500}) with pentetic acid (PA) for the generation of PA@Pal_{500} nanosorbent. The novelty of this work is mainly correlated to (i) the assembled PA@Pal_{500} is a novel nanosorbent and not previously prepared, published, or employed in removal studies. (ii) PA@Pal_{500} was synthesized from an eco-friendly pentetic acid (PA). (iii) A simple, low-cost, and green microwave-assisted synthesis in a one-step route to functionalize palygorskite with PA for introducing some reactive surface sites and groups to favor capturing and binding with Cd(II). (iv) The green microwave irradiation process required only 15 min without using any organic or hazardous solvents, as reported and compared to other publications. (v) Effective Cd(II) removal at low/trace concentrations combined with retained performance under high ionic strength is aimed to represent a regime relevant to real and saline water matrices in which many high-capacity adsorbents lose selectivity. (vi) The spent PA@Pal_{500} was confirmed as an efficient regenerable nanosorbent for numerous recycling conditions. (vii) Combined experimental–computational (DFT/Monte Carlo) analysis that provides molecular-level insight into the preferred Cd(II) binding sites, beyond the purely empirical characterization common in comparable studies. Moreover, the chemical structure and morphology of Pal_{500} are aimed to be confirmed by several approaches such as FTIR, SEM, EDX, XRD, and BET. This study is also directed to explore how key experimental factors such as pH, initial concentration of Cd(II), temperature, dosage, reaction time, and ionic strength affect the efficiency of Cd(II) removal. In addition, the regeneration capability of PA@Pal_{500} is also explored to enhance its long-term performance, aiming to improve cadmium removal from water and promote a more sustainable treatment approach. To better understand how Cd(II) ions interact with both Pal_{500} and PA, computational studies were carried out via combined DFT calculations with Monte Carlo simulations. Using the DMol³ module, the optimized molecular geometries were analyzed with the electronic properties, including electrostatic potential surfaces, total density of states (TDOS), and Fukui functions. In addition, Monte Carlo simulations employing the Adsorption Locator annealing method were used to identify the most stable adsorption arrangements with their corresponding energies. Together, these results provided insight into the binding behavior of Cd(II), highlighting the contrast between the surface adsorption on Pal_{500} and its more localized coordination with PA.

Results and discussion

Characterization

FTIR confirmation technique was used to examine the surface functionalities on PA@Pal_{500} nanosorbent and to clarify how it interacts with Cd(II)

ions as shown in Fig. 1. The spectra collected before and after Cd(II) adsorption display clear vibrational features from both Pal_{500} and PA, confirming successful surface modification and effective Cd(II) binding at 3452 , 2933 , 1623 , 1409 , 1188 , 1030 cm^{-1} . The broad band centered at 3452 cm^{-1} is correlated to the vibrations of hydroxyl groups originating from residual structural $-\text{OH}$ of Pal_{500} and H-bonded hydroxyl groups of PA. The band observed at 2933 cm^{-1} refers to C–H stretching of methylene ($-\text{CH}_2-$) in PA, confirming the successful immobilization of the organic ligand onto the Pal_{500} surface³⁶. A prominent band at 1623 cm^{-1} is mainly assigned to the asymmetric stretching vibrations of carboxylate ($-\text{COO}^-$) groups of PA, with a possible contribution from the bending mode of adsorbed water. The band at 1409 cm^{-1} is associated with symmetric stretching of carboxylate groups and also shows noticeable changes following Cd(II) uptake³⁷. The absorption band at 1188 cm^{-1} is combined with C–N stretching and C–O vibrations of PA overlapping with Si–O vibrations of Pal_{500} . The band at 1030 cm^{-1} is attributed to (Si–O–Mg) and (Si–O–Al) of the stretching (Si–O) in Pal_{500} with notable intensity, as previously demonstrated²⁹. The low-frequency bands appearing in the range of 520 – 473 cm^{-1} are assigned to Si–O bending vibrations and lattice modes of Pal_{500} . To reveal clear and diagnostic differences after Cd(II) uptake (Fig. 1a), the asymmetric carboxylate ($-\text{COO}^-$) stretch of PA was shifted from 1626.6 to 1589.8 cm^{-1} ($\Delta\nu = -37 \text{ cm}^{-1}$), while the symmetric stretch was essentially unchanged ($1409 \rightarrow 1410 \text{ cm}^{-1}$). The resulting narrowing in the ($\nu_{\text{asym}} - \nu_{\text{sym}}$) from 217 to 180 cm^{-1} identifies the deprotonated carboxylate oxygen as the principal Cd(II)-interacting group. In parallel, the $-\text{OH}/-\text{COOH}$ band near 3450 cm^{-1} were also confirmed to develop an additional feature ($\sim 3541 \text{ cm}^{-1}$), and the C–O/C–N region ($\sim 1188 \text{ cm}^{-1}$) showed minor intensity changes, consistent with the participation of oxygen-bearing groups in Cd(II) binding. In the low-frequency fingerprint region, Cd–O lattice vibrations overlapped with the strong Si–O framework bands and are therefore masked to partly account for the apparent similarity of the two spectra in that range. Taken together, these shifts, particularly the carboxylate bands, indicated that Cd(II) interacts predominantly through electrostatic attraction to the negatively charged $-\text{COO}^-$ and $-\text{OH}$ groups, with only a minor localized coordination contribution, and this is in agreement with the physical adsorption mechanism inferred from the Dubinin–Radushkevich mean free energy ($<8 \text{ kJ/mol}$)³⁷.

The SEM image of PA@Pal_{500} (Fig. 1b) reveals a rough, heterogeneous surface composed of aggregated particles with an open, porous texture. The micrograph displays the characteristic fibrous (lath/needle-like) crystals of palygorskite, partially organized into bundles and intergrown with platy aggregates, indicating that the fibrous framework of the clay is retained after calcination and PA functionalization. The resulting interparticle voids and irregular morphology provide an accessible, high-surface-area network that favors Cd(II) adsorption. No large, separate crystalline phases are evident on the surface, consistent with PA being dispersed over the clay rather than forming an isolated segregated phase, as further supported by the FT-IR (and EDX) results³⁰.

EDX analysis (Fig. 1c) was used to determine the elemental composition of the PA@Pal_{500} nanosorbent. The spectrum showed that oxygen was the dominant element ($54.38 \pm 0.62 \text{ mass } \%$ and $60.86 \pm 0.70 \text{ atom } \%$), reflecting the abundance of oxygen-containing groups from both PA and the silicate framework of palygorskite. Silicon appeared at $18.65 \pm 0.31 \text{ mass } \%$, confirming that the clay silicate structure remained intact following thermal treatment at 500°C . The detection of carbon ($9.88 \pm 0.31 \text{ mass } \%$) and nitrogen ($2.65 \pm 0.23 \text{ mass } \%$) provides strong evidence of successful functionalization with PA. Additionally, minor amounts of Na, Mg, Al, K, Ca, and Fe correspond to naturally occurring elements in palygorskite and may play supplementary roles in metal adsorption. Overall, the EDX results verify the acceptable incorporation of PA into the Pal_{500} matrix and support the PA@Pal_{500} 's suitability for efficient Cd(II) ion adsorption^{28,29}.

The XRD pattern of the PA@Pal_{500} nanosorbent (Fig. 1d) shows both crystalline and amorphous characteristics, reflecting the structural changes caused by heating and surface functionalization at $2\theta = 8.25^\circ$, 17.98° , 19.46° , 26.15° , and 34.67° . The distinct 2θ peak at 8.25° is based on the typical

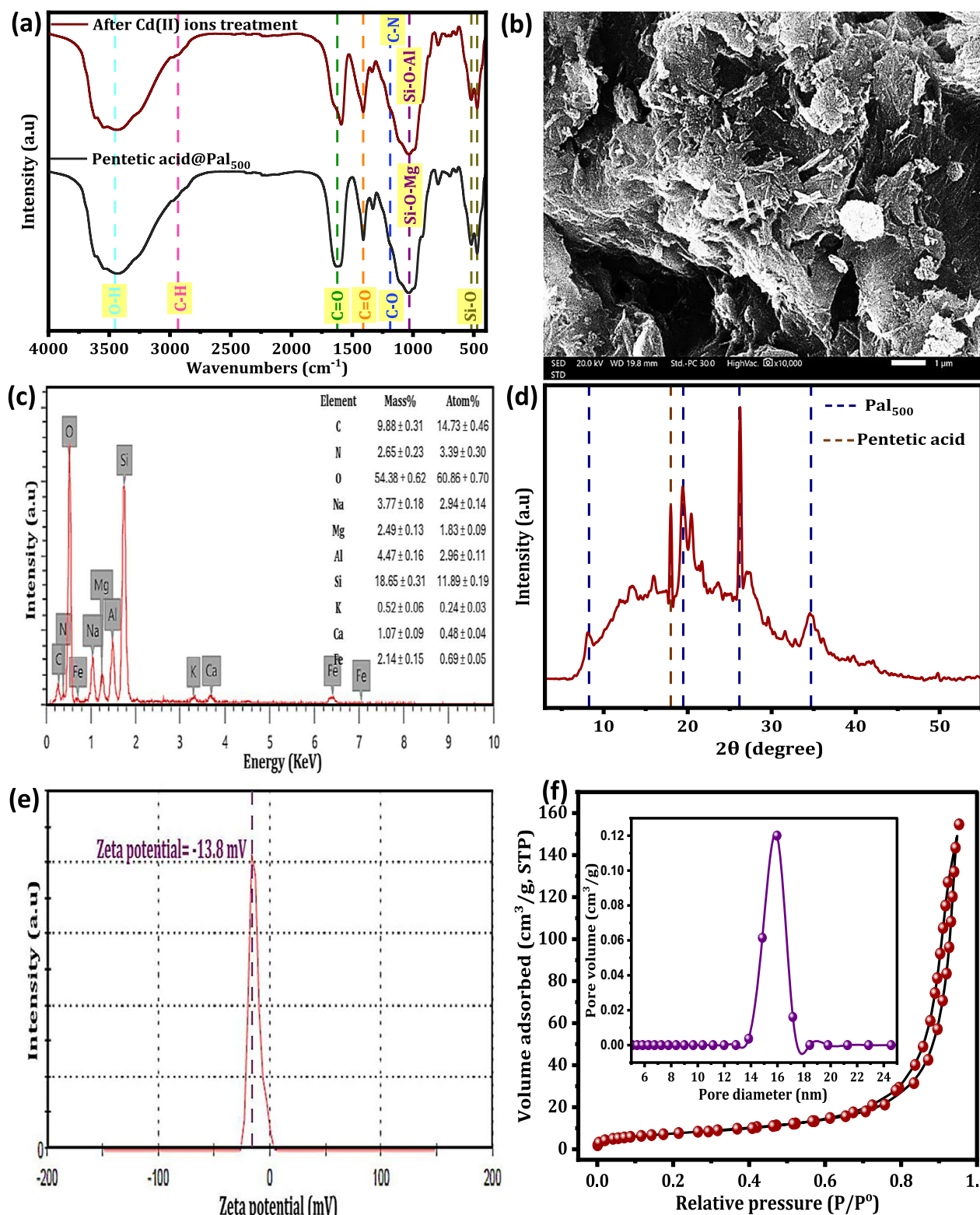


Fig. 1 | Characterization of the PA@Pal₅₀₀ nanosorbent. a FTIR spectra before and after Cd(II) adsorption, **b** SEM micrograph, **c** EDX spectrum and elemental composition analysis, **d** XRD pattern, **e** zeta potential distribution, and **f** N₂ adsorption-desorption isotherms.

diffraction of Pal₅₀₀, confirming that the clay silicate framework remains largely intact even after calcination at 500 °C³⁸. Likewise, the strong peak at $2\theta = 19.46^\circ$ represents a well-defined crystalline phase of Pal₅₀₀, indicating that the mineral fibrous structure survives the high-temperature treatment³⁹. The peak at $2\theta = 26.15^\circ$ suggests some degree of structural modification resulting from the interaction between PA and the Pal₅₀₀

surface. This may be owed to the formation of a new crystalline arrangement or a shift in the existing crystal order due to functionalization. The diffraction at $2\theta = 17.98^\circ$ indicates slight disruption within the Pal₅₀₀ lattice, likely caused by the incorporation of PA, which altered the clay surface environment. In addition, the peak at $2\theta = 34.67^\circ$ supports the presence of remaining crystalline domains within the composite, showing that although

functionalization introduces some amorphous characteristics, significant crystallinity is still preserved⁴⁰. Overall, the XRD analysis demonstrates that the PA@Pal₅₀₀ nanosorbent contains both crystalline and amorphous phases. Thermal treatment and surface modification influence the degree of structural order in Pal₅₀₀, while enhancing its surface properties for effective Cd(II) ion adsorption. Finally, the fibrous crystals exhibited an approximate particle size in the range of 35–44 nm, in agreement with the nanoscale fibrous morphology of palygorskite

The zeta potential measurement of the PA@Pal₅₀₀ nanosorbent (Fig. 1e) provides insight into its surface charge and stability in suspension. The PA@Pal₅₀₀ shows a zeta potential of −13.8 mV, indicating a negatively charged surface. This level of charge suggests that the particles have moderate colloidal stability, as the negative charge helps them repel one another and reduces excessive aggregation. The negative surface charge is likely due to the functional groups introduced by PA, particularly carboxylate and amine groups, which interact with the Pal₅₀₀ surface. The zeta potential of 5.01 mV indicates some variability in particle surface charge, probably influenced by environmental factors, viz pH changes or the ionic strength of the dispersion medium. The measured conductivity of 1.06 mS cm^{−1} reflects the existing ions in solution. These ions can compress or alter the electrical double layer around the particles, which in turn can affect the measured zeta potential. Overall, the negative surface charge and moderate colloidal stability of the PA@Pal₅₀₀ nanosorbent make it well-suited for adsorption processes, particularly in capturing positively charged metal ions or contaminants. The zeta potential results confirm that the PA@Pal₅₀₀ maintains sufficient stability in suspension, which is essential for efficient Cd(II) ion adsorption performance³⁰.

The N₂ adsorption–desorption analysis of the PA@Pal₅₀₀ nanosorbent reveals key insights into its pore structure and surface properties. The sample shows a characteristic type IV isotherm accompanied by H3 hysteresis loop features typically associated with mesoporous materials (Fig. 1f). This behavior indicates the presence of a pore diameter corresponding to 15.96 nm, confirming that the PA@Pal₅₀₀ contains well-developed mesopores that play an important role in its adsorption performance⁴⁰. The appearance of an H3-type hysteresis loop (Fig. 1f) in the desorption branch further reinforces this conclusion. Such loops are commonly observed in materials made up of plate-like particles that pack together to form slit-shaped or irregular mesoporous structures. These structural features tend to trap N₂ molecules during desorption, which explains the observed irreversible adsorption. In terms of textural properties, the PA@Pal₅₀₀ exhibits a BET surface area of 28.3 m²/g and a total pore volume of 0.11 cm³/g. These values reflect a substantial available surface and significant pore capacity, both of which are essential for efficient adsorption⁴¹. Overall, the mesoporous architecture enhances the PA@Pal₅₀₀ ability to capture Cd(II) ions and numerous contaminants, highlighting its suitability for environmental remediation and other adsorption-based applications.

Batch experiment optimization of Cd(II) removal by PA@Pal₅₀₀ nanosorbent

Palygorskite is a hydrated magnesium–aluminum silicate with a distinctive fibrous (needle-like) morphology and a 2:1 inverted-ribbon structure containing structural and zeolitic channels. This architecture provides abundant surface silanol (Si–OH) and structural oxygen groups, a permanent negative structural charge with exchangeable cations, and good chemical, thermal, and mechanical stability, in addition to its natural abundance and low cost. These features make palygorskite an attractive and inexpensive substrate for effective binding with heavy metal cations such as Cd(II). However, the density and selectivity of its native binding sites are limited, which motivates surface functionalization to enhance metal uptake. On the other hand, pentetic acid (PA, diethylenetriaminepentaacetic acid) was selected as the functionalizing ligand due to a high density of oxygen- and nitrogen-donor sites (five carboxyl and three amino groups) with a well-documented strong affinity for divalent metal cations. Grafting PA onto palygorskite clay can afford several advantageous features, such as (i) incorporation of a large number of electron-rich carboxylate and amine

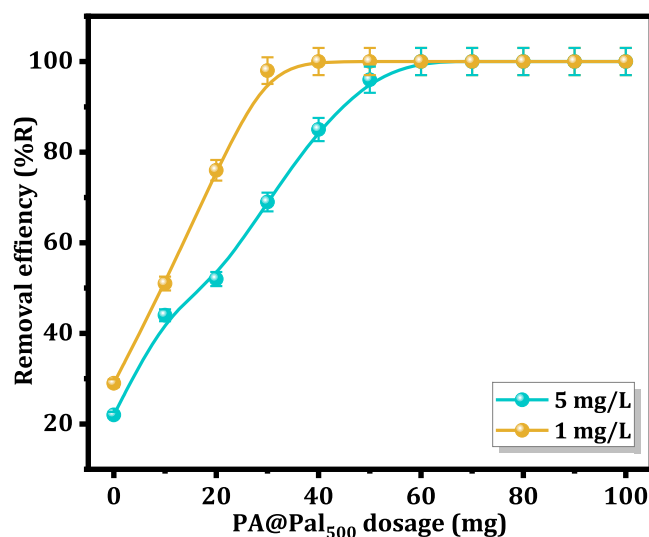


Fig. 2 | Effect of PA@Pal₅₀₀ nanosorbent dosage on the removal efficiency of 1 and 5 mg/L Cd(II) ions. The amount of PA@Pal₅₀₀ nanosorbent used in this factor exhibited a clear influence on how effectively Cd(II) ions were removed by using a 30 min shaking time.

groups, (ii) increment of the negative surface charge that drives the electrostatic attraction and complexation with Cd(II) ions, and (iii) combination of the metal affinity by the PA ligand with the stability of clay support. To our knowledge, PA has not previously been immobilized on palygorskite for Cd(II) removal, which forms the basis of the present study.

The amount of PA@Pal₅₀₀ nanosorbent used exhibited a clear influence on how effectively Cd(II) ions were removed, as shown in Fig. 2 using a 30 min shaking time. When the dosage was improved from 10 to 40 mg, the removal efficiency of 1 mg/L Cd(II) solution rose from 51% to 100%, after which it remained steady. For the 5 mg/L concentration, the efficiency was similarly enhanced from 44% to 100% as the dosage went from 10 mg to 60 mg, with no further improvement beyond 60 mg. Overall, higher dosages consistently produced better removal performance, mainly due to the greater number of available adsorption sites⁴². In contrast, when no PA was added (0 mg dosage), the removal efficiencies were lower: 29% and 22%, for 1.0 and 5.0 mg/L, respectively. These results highlight that the modification of Pal₅₀₀ with PA increases both the reactivity and activity of binding sites on the surface, enhancing overall adsorption performance⁴³.

It is well known that exchangeable cations such as Na⁺, K⁺, Ca²⁺, and a small fraction of Mg²⁺ occupy the structural channels and surface sites of palygorskite and therefore are available to favor cation exchange with Cd(II) ions. The structural Al³⁺, bulk Mg²⁺, and Fe³⁺ are held within the octahedral framework and are therefore only released under strongly acidic conditions, if the silicate structure is dissolved or decomposed. During Cd(II) adsorption at the working pHs, charge neutrality was maintained by displacement of the loosely held channel cations into solution. Consistent with this, the framework cations of Al and Fe are not expected to be released. However, ion exchange of channel cations could be operated alongside surface complexation and electrostatic interaction. This indicates that Cd(II) uptake was chiefly accomplished by cation exchange of weakly bound cations rather than dissolution of the clay framework.

The pH of the metal ion solution plays a crucial role in determining the chemical forms of Cd(II) ions and the surface charge of PA@Pal₅₀₀. Figure 3 noticeably refers to the variation in RE % for Cd(II) across the tested pH range (1–7). The PA@Pal₅₀₀ performed best at neutral pH, particularly between pH 7. In contrast, performance at pH 1 was significantly lower (10% and 6% for 1 and 5 mg/L, respectively), likely due to the severe acidity, which can damage the surface structure and deactivate functional adsorption sites on PA@Pal₅₀₀ via protonation⁴⁴. This can be understood by considering the competition of hydrogen (H⁺) vs Cd(II) for available binding

sites. At low pH values (below pH 4), high H^+ ion concentration resulted in protonating the amine groups in PA. This protonation minimized the number of accessible binding sites for Cd(II) and also created a positively

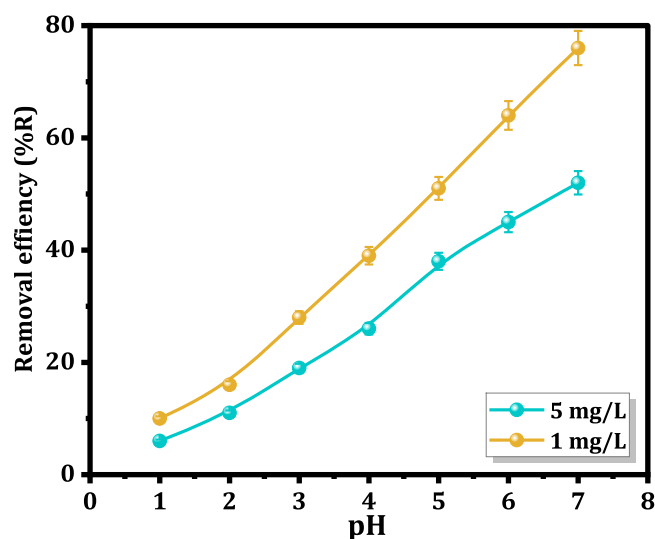


Fig. 3 | Effect of solution pH on the removal efficiency of 1 and 5 mg/L Cd(II) ions by PA@Pal₅₀₀ nanosorbent. It refers to the variation in RE % for Cd(II) across the tested pH range (1–7), with the best performance at neutral, particularly between pH 7.

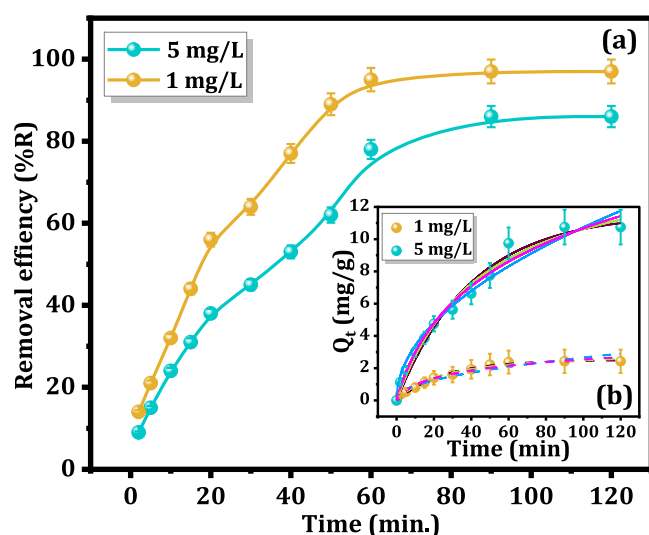


Fig. 4 | Illustration for the impact of shaking time on the removal efficiency (%) of 1 and 5 mg/L Cd(II) ions by the variation in time of reaction from 2 to 120 min, while keeping all other conditions constant. a Adsorption dynamics over time to **b** adsorption isotherms for Cd(II) removal onto PA@Pal₅₀₀ nanosorbent at 1 mg/L (yellow) and 5 mg/L (cyan) concentrations.

charged surface, which electrostatically repels Cd(II) ions and thus ultimately lowers adsorption efficiency. As the pH increases above 4, the amine groups become deprotonated, fewer protons compete for the sites, and the surface becomes more negatively charged. These sites were more readily attracted and coordinated with Cd(II) ions, enhancing both electrostatic attraction and chelation. The highest adsorption performance occurred around pH 6–7, where the deprotonated amine groups are most effective at binding Cd(II)⁴⁵. However, extremely acidic or highly alkaline conditions can lead to the dissolution of metallic species from the Pal₅₀₀ framework into the solution. Such dissolution compromises both the structural integrity and the long-term stability of PA@Pal₅₀₀. For this reason, the pH of the solution was carefully controlled and maintained close to neutral conditions (around pH 6) during the experiments. This ensured optimal removal efficiency, while avoiding Cd(II) precipitation and protecting the structural stability of PA@Pal₅₀₀⁴⁶.

Kinetic study of Cd(II) ions adsorption onto PA@Pal₅₀₀ nanosorbent

Figure 4a illustrates the impact of shaking time on the removal efficiency (%) of Cd(II) ions. To explore this factor, the time was varied from 2 to 120 min, while keeping all other conditions constant, viz temperature at 298 K, pH 6, 20 mg dose of PA@Pal₅₀₀, and 1 and 5 mg/L Cd(II). The findings reveal that the removal efficiency rapidly increased during the initial stages, indicating that adsorption was most effective at the beginning⁴⁷. As the process approached equilibrium, the removal rate was stabilized at 95% for 1 mg/L and 86% for 5 mg/L, with maximum adsorption achieved around 60 min and 90 min, respectively.

The kinetic modeling can reveal distinctive adsorption mechanisms, while the data could be generally better evaluated by application of the PFO, PSO, Intraparticle, and Elovich models (Table S3), as evidenced by a higher R^2 coefficient as presented in Table 1. The pseudo-first-order (PFO) can prove advantageous for studying Cd(II) adsorption onto PA@Pal₅₀₀. Its main strength lies in its simplicity and clear interpretation of the data, while the adsorption rate is directly linked to the number of available active sites (Table 1). The rate constants, $K_1 = 0.039 \text{ min}^{-1}$ for 1 mg/L and 0.025 min^{-1} for 5 mg/L, indicate that adsorption occurs more rapidly at lower concentrations. This suggests strong bonding of Cd(II) ions to the PA@Pal₅₀₀ surface. The model offers a practical way to evaluate removal efficiency and optimize process conditions. Furthermore, the experimental equilibrium adsorption capacities ($Q_e(\text{exp}) = 2.42$ and 10.75 mg/g) closely match the calculated values ($Q_e(\text{calc}) = 2.49$ and 11.57 mg/g), and are supported by high correlation coefficients ($R^2 = 0.991$ and 0.982). Such results confirm the reliability of PFO in the interpretation of the adsorption kinetics by providing valuable insights for improving system performance⁴⁸.

The pseudo-second-order (PSO) offers significant advantages for analyzing Cd(II) adsorption onto PA@Pal₅₀₀. With rate constants $K_2 = 0.014$ and 0.0014 min^{-1} and calculated equilibrium capacities ($Q_e(\text{calc}) = 3.08$ and 15.48 mg/g) for 1 and 5 mg/L Cd(II), respectively, the model suggests that adsorption efficiency depends on the square of the number of active sites. This often indicates that the rate-determining step (RDS) is controlled by chemical interactions rather than physical processes. As shown in Fig. 4b, the PSO model predicts higher adsorption capacities, supported by strong correlation coefficients ($R^2 = 0.993$ and 0.983). Compared to other kinetic models, PSO demonstrates superior accuracy in estimating Q_e values, making it highly relevant for

Table 1 | Non-linear adsorption isotherm models and parameters for adsorption of Cd(II) ions with standard deviations within ± 0.2 – 2.5%

Isotherm model	Langmuir				Freundlich			D-R			Temkin			
Plot	Q_e vs C_e				Q_e vs C_e			Q_e vs C_e			Q_e vs C_e			
Constant	q_{max}	b	R^2	R_L	n	K_f	R^2	Q_{DR}	K_{ad}	R^2	E	b_T	R^2	a_T
Unit	mg g^{-1}	L mg^{-1}	-	-	-	L mg^{-1}	-	mg g^{-1}	$\text{mol}^2 \text{J}^{-2}$	-	J mol^{-1}	J mol^{-1}	-	L mg^{-1}
20 mg	3.26	-1.33	-1.17	(-0.093) to (-5.88)	1.36	6.87	0.988	8.74	2.63	0.839	435.68	956.55	0.945	4.34
50 mg	7.59	5.87	0.94	0.017–0.145	3.86	5.99	0.982	1	1	-7.96	707	2477.6	-11.50	1

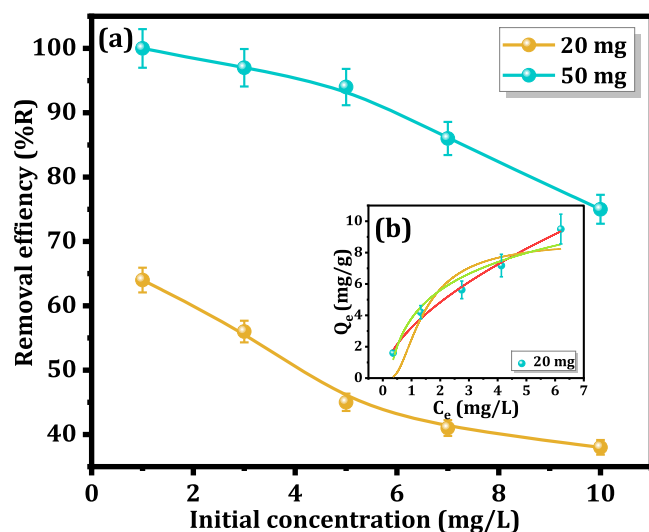


Fig. 5 | Complete removal of 1 mg/L Cd(II) by PA@Pal₅₀₀, and once the concentration surpasses a critical level (5 mg/L), the active sites become saturated, causing the adsorption efficiency to level off. a Effect of initial concentration (mg/L) on the removal efficiency of Cd(II) ions by PA@Pal₅₀₀. **b** Adsorption isotherms for Cd(II) removal by PA@Pal₅₀₀ nanosorbent with experimental data points (Langmuir, Freundlich, Temkin, and D-R).

designing and optimizing water treatment systems where precise control of efficiency and capacity is essential. However, Table 1 reveals that Cd(II) removal aligns more closely with the PFO model, as the experimental Q_e values correspond better to the PFO-calculated values than to those predicted by PSO. This suggests that while PSO provides valuable insights into chemical interaction mechanisms, PFO remains a more representative model for this adsorption process⁴⁹.

The intraparticle diffusion model provides valuable uptake insights of Cd(II) by PA@Pal₅₀₀ by highlighting the role of diffusion in the overall removal process. For Cd(II) at 1 and 5 mg/L, the model parameters include K_{diff} values of 0.25 and 1.07 mg/g min^{1/2} and C values of 0.107 and 9.99×10^{-27} mg/g, respectively. These results suggest that diffusion into the material internal cavities occurs at a relatively slow rate, as indicated by the low K_{diff} values, while the PA@Pal₅₀₀ nanosorbent still exhibits strong sorption capacity for Cd(II). The C values reflect the boundary layer effect, confirming that surface adsorption plays a significant role alongside intraparticle diffusion⁵⁰. One of the model's key strengths is its ability to separate the adsorption process into distinct phases, surface sorption and internal diffusion, providing a clearer understanding of the mechanisms involved. This insight is crucial for improving the nanosorbent pore structure and surface properties, ultimately enhancing its performance in environmental applications.

The Elovich kinetic model provides important insights into the Cd(II) capture by PA@Pal₅₀₀, particularly for heterogeneous surfaces where chemical interactions dominate. For initial Cd(II) concentrations of 1 and 5 mg/L, the model parameters were $\beta = 1.24$ and 0.217 and $\alpha = 0.179$ and 0.420 mg/g min, respectively (Fig. 4b). This model effectively describes the exponential decline in adsorption rate over time as surface coverage increases and binding of molecules becomes highly significant. The relatively low β value indicates a substantial removal efficiency enhancement vs time, suggesting that PA@Pal₅₀₀ maintains a high adsorption capacity (Q_e) even at elevated coverage levels⁵¹. The flexibility of the Elovich model makes it particularly useful for understanding adsorption on heterogeneous surfaces, offering valuable guidance for improving the nanosorbent properties and enhancing its performance in practical water treatment applications.

The kinetics analysis showed that Cd(II) adsorption by PA@Pal₅₀₀ followed the PFO model most closely, as indicated by the strong agreement of predicted capacities and experimental. The adsorption rate was faster at lower initial metal concentrations, reflecting the greater availability of active

sites under these conditions. Although the PSO model also provided a good fit, indicating chemical interaction role. Moreover, the intraparticle diffusion and Elovich models revealed that adsorption took place through multiple stages. These findings suggest that the process involves both surface interactions and intraparticle diffusion within a heterogeneous nanosorbent structure.

As the Cd(II) ion concentration increases, the removal efficiency gradually declines. Interestingly, PA@Pal₅₀₀ can still achieve complete removal (100%) of Cd(II) by 1 mg/L (Fig. 5a). At lower concentrations, the material's abundant active sites enable highly effective adsorption. However, once the concentration surpasses a critical level (around 5 mg/L), these sites become saturated, causing the adsorption efficiency to level off. This saturation means that a smaller proportion of Cd(II) ions can be captured per unit of PA@Pal₅₀₀⁵².

Additionally, the interaction between PA@Pal₅₀₀ nanosorbent and Cd(II) ions was analyzed using various adsorption isotherm models. These models help in explaining how the material adsorbs Cd(II) ions and provide key constants that indicate the amount of PA@Pal₅₀₀ needed to capture a given quantity of ions. To interpret the experimental adsorption data, four models were applied, viz Langmuir, Freundlich, Dubinin–Radushkevich (D–R), and Temkin⁵³. The Langmuir isotherm, which assumed monolayer adsorption onto a uniform surface with a limited number of binding sites, did not accurately describe Cd(II) removal in this study based on the lower R^2 values obtained for 20 and 50 mg. The calculated maximum capacities (q_{max}) were 3.26 mg/g for 20 mg and 7.59 mg/g for 50 mg. The notably high q_{max} for Cd(II) suggests that PA@Pal₅₀₀ exhibits a strong affinity for these ions, mainly due to electrostatic interactions and complexation with carboxylic groups on the PA surface. The separation factor (R_L), which indicates adsorption feasibility, ranged from -0.093 to -5.88 for 20 mg and from 0.017 to 0.145 for 50 mg. For 50 mg, the R_L values fell within the favorable range ($0 < R_L < 1$), confirming efficient adsorption. Overall, these results show that Cd(II) removal didn't follow the Langmuir model⁵⁴.

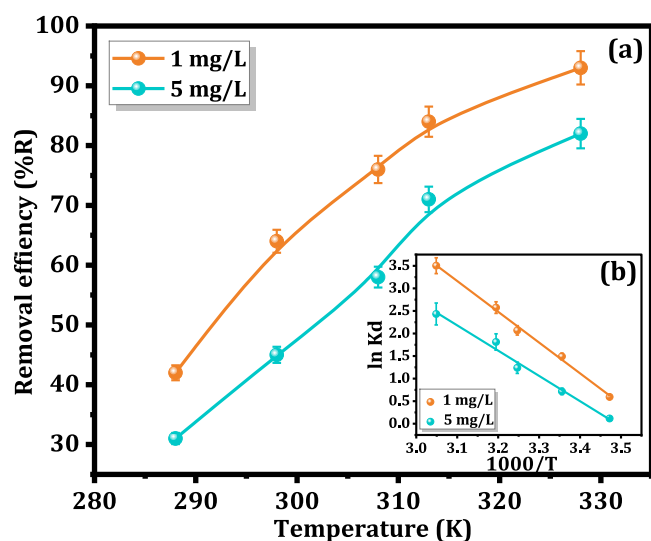
The Freundlich isotherm accounted for adsorption on heterogeneous surfaces to form multilayer interactions and provided a better match than the Langmuir model. This was evident from the high correlation coefficients: $R^2 = 0.988$ for 20 mg and $R^2 = 0.982$ for 50 mg. The Freundlich constants (K_F) were computed to refer to 6.87 mg/g for 20 mg and 5.99 mg/g for 50 mg as the related capacity. The n values (1.36 and 3.86) indicate favorable adsorption, as $0 < 1/n < 1$ reflects strong surface affinity. The higher K_F value suggests a greater tendency for multilayer adsorption, driven by interactions of Cd(II) ions and the PA@Pal₅₀₀ surface. Overall, the Freundlich model outperformed the Langmuir model, as shown by its superior R^2 values, while also capturing the heterogeneity of the adsorption sites (Fig. 5b and Table 2)⁵⁵.

The D–R model can provide important parameters for Cd(II) uptake by PA@Pal₅₀₀, including adsorption capacities ($Q_{DR} = 8.74$ mg/g for 20 mg and 1.00 mg/g for 50 mg), adsorption constants ($K_{ads} = 2.63$ for 20 mg and 1.00 for 50 mg, mol²/J²), and mean free energy ($E = 435.68$ J/mol for 20 mg and 707 J/mol for 50 mg) (Fig. 5b). This model is particularly useful because it differentiates between physical and chemical adsorption⁵⁶. Based on the computed data, the D–R model did not adequately describe the equilibrium results owing to the low fittings ($R^2 = 0.839$) and ($R^2 = -7.96$) by using 20 and 50 mg dosages of PA@Pal₅₀₀, respectively. Therefore, the D–R model could be unacceptable for interpretation of mechanistic reactions. The predominantly electrostatic and/or cation exchange nature of Cd(II) uptake is consistent with the energetically heterogeneous surface indicated by the Freundlich isotherm.

The Temkin isotherm (Fig. 5b), which considers the distribution of binding energies during adsorption, showed a weak R^2 correlation coefficient for both 20 and 50 mg indicating model inadequately to describe the adsorption process. The heat of adsorption (b_T) increased from 956.55 J/mol at 20 mg to 2477.6 J/mol at 50 mg, suggesting stronger interactions at a higher PA@Pal₅₀₀ dose. In contrast, the adsorption affinity (a_T) was much higher for 20 mg (4.34 L/g) compared to 50 mg (1.00 L/g), implying that PA@Pal₅₀₀ performs better at lower nanosorbent amounts when active sites

Table 2 | Thermodynamic parameters for adsorption of Cd(II) ions with standard deviations within ± 0.8 –2%

Cd(II)	Temp. (K)	K_d (L.g ⁻¹)	R^2	Thermodynamic parameters		
				ΔG° (KJ.mol ⁻¹)	ΔH° (KJ.mol ⁻¹)	ΔS° (J mol ⁻¹ .K ⁻¹)
1 mg/L	288	1.81	1	−1.42	56.72	202.13
	298	4.44		−3.70		
	308	7.92		−5.30		
	313	13.13		−6.70		
	328	33.21		−9.55		
5 mg/L	288	1.12	0.987	−0.28	46.60	162.59
	298	2.05		−1.77		
	308	3.45		−3.17		
	313	6.12		−4.71		
	328	11.39		−6.63		

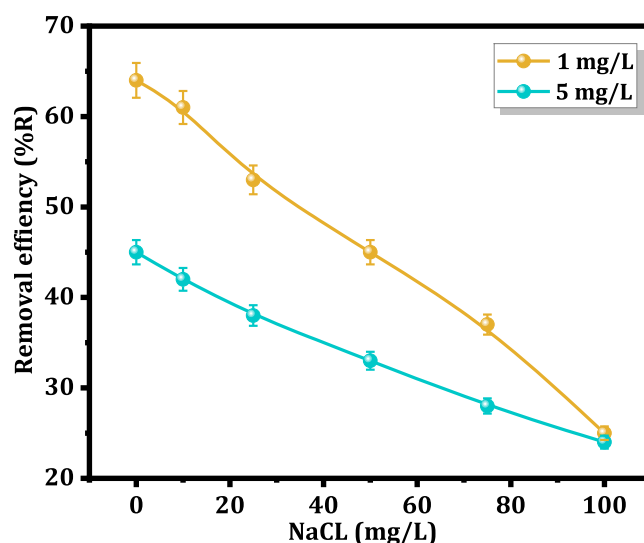
**Fig. 6 | Improvement in the removal efficiency with temperature increase indicates an endothermic behavior due to higher kinetic energy of Cd(II) ions at elevated temperatures to favor enhanced and promoted diffusion and adsorption toward the PA@Pal₅₀₀ surface. a** Effect of temperature on removal efficiency was (b) Van't Hoff linear plot of $\ln K_d$ vs $1000/T$ for Cd(II) adsorption onto PA@Pal₅₀₀ nanosorbent.

are less saturated. These findings highlight the complexity of Cd(II) adsorption on PA@Pal₅₀₀ and suggest that, while the Temkin model offers some insight into binding behavior, it does not fully capture the adsorption mechanism⁵⁷.

Among the four evaluated isotherm models, the Freundlich model provided the best description of Cd(II) adsorption onto PA@Pal₅₀₀, indicating multilayer production on heterogeneous active sites. Moreover, the Temkin model further supported the presence of strong interactions between Cd(II) ions and PA@Pal₅₀₀, highlighting the role of varying binding energies in the adsorption process. Collectively, these findings demonstrate the strong potential of PA@Pal₅₀₀ nanoparticles as an efficient nanosorbent for heavy metal treatment applications.

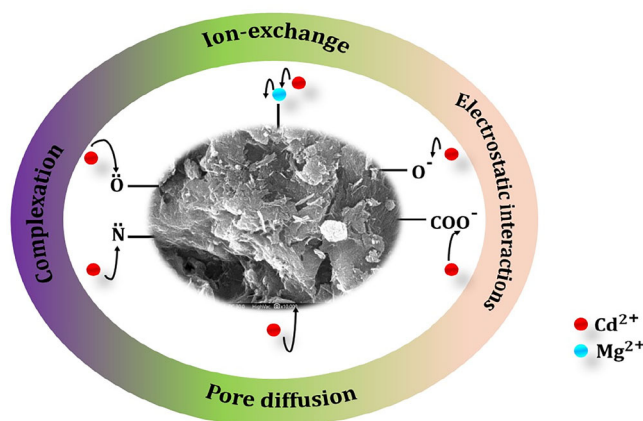
The influence of temperature on Cd(II) ion removal was examined across a temperature range from 288 K to 328 K. The results reveal that removal efficiency was improved by temperature increase, indicating an endothermic. At 288 K, an initial concentration of 1 mg/L achieved a RE% of 42%, while 5 mg/L reached 31%, as illustrated in Fig. 6a. Such temperature-dependent increase can be attributed to the higher kinetic energy of Cd(II) ions at elevated temperatures, which enhances their diffusion toward the PA@Pal₅₀₀ surface and promotes adsorption⁵⁸.

Thermodynamic parameters from the van't Hoff plot (Fig. 6b and Table 2) gave negative ΔG° (−1.42 to −9.55 kJ/mol for 1 mg/L and −0.28 to −6.63 kJ/mol for 5 mg/L), positive ΔH° (56.72 and 46.60 kJ/mol), and positive ΔS° (202.13 and 162.59 J/mol.K). The negative ΔG°

**Fig. 7 | Effect of ionic strength on adsorption of Cd(II) onto PA@Pal₅₀₀ nanosorbent under numerous ionic strengths using solutions of NaCl (0 to 100 mg/L). When the Na⁺ ion reached 25 mg/L, the RE % by PA@Pal₅₀₀ decreased by only 11% for 1 mg/L Cd(II) and 7% by 5 mg/L Cd(II), to refer to low contribution to removal effectiveness.**

confirms that the adsorption is spontaneous, and its small magnitude (<10 kJ/mol) is consistent with a predominantly physical interaction⁴⁴. The coexistence of an endothermic process ($\Delta H^\circ > 0$) and spontaneity ($\Delta G^\circ < 0$) is explained through $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$: because ΔS° is positive and large, the entropic term ($-T\Delta S^\circ$) is sufficiently negative to outweigh the positive ΔH° , giving $\Delta G^\circ < 0$. The process is therefore entropy-driven, and its spontaneity increases with temperature, in agreement with the observed rise in removal efficiency. The positive ΔH° and large positive ΔS° are attributed to desolvation: prior to binding, both the Cd(II) ions and the PA@Pal₅₀₀ surface are hydrated, and adsorption requires partial removal of these ordered hydration shells (an endothermic step), while the accompanying release of water molecules and displaced exchangeable cations into the bulk solution increases the overall disorder of the system (positive ΔS°)⁵⁹. Accordingly, the positive ΔH° reflects the net energy cost of desolvation associated with a predominantly electrostatic/physical adsorption rather than the formation of strong chemical bonds, consistent with the low Dubinin-Radushkevich mean free energy (<8 kJ/mol) and the small negative ΔG° values⁶⁰.

The Cd(II) adsorption under numerous ionic strengths was assessed using solutions of NaCl (0–100 mg/L). When the Na⁺ ion reached 25 mg/L the RE % by PA@Pal₅₀₀ decreased by only 11% for 1 mg/L Cd(II) and 7% for 5 mg/L Cd(II), and this declining trend is negligible for effective removal (Fig. 7). Even at a 100-fold increase in Na⁺ ions, the adsorption efficiency



Scheme 1 | Mechanism of PA@Pal₅₀₀ nanosorbent in the adsorption of Cd(II) from water.

remained around 25% and 24%, respectively. These findings highlight the selectivity of PA@Pal₅₀₀ for Cd(II) ions under high ionic strength environments, underscoring its practical applicability. The slight decline in adsorption could be attributed to direct competitive interference from excess Na⁺ ions, which partially disrupt the ion-exchange mechanism⁶¹.

The regeneration ability of PA@Pal₅₀₀ was assessed to explore its suitability in repeated Cd(II) removal cycles. Among several tested eluents, 0.01 M HCl proved to be the most effective reagent for desorption. The process was carried out for 30 min at 298 K using 5 mg/L initial Cd(II) concentration, and the resulting data are illustrated in Fig. S1. The regeneration and reusability of PA@Pal₅₀₀ were evaluated over nine consecutive adsorption–desorption cycles using 0.01 M HCl as the eluent (30 min, 298 K, 5 mg/L Cd(II)). As shown in Fig. S1, the adsorption efficiency decreased from 96% for the fresh sorbent to 88, 79, 68, 59 and 51% over the first five cycles, and then stabilized, declining only marginally to 48, 45 and 42% in the sixth, seventh and ninth cycles, respectively; the sorbent thus retained about 42% of its initial efficiency after nine cycles. The relatively steep decline over the first five cycles is attributed to sorbent loss during the washing and separation steps, to incomplete desorption of Cd(II) from the stronger binding sites, and to a gradual loss of accessible functional groups under repeated acidic elution. The subsequent plateau (cycles 5–9) indicates that the remaining sites are comparatively stable and reusable. Overall, PA@Pal₅₀₀ can be regenerated and reused over multiple cycles with a stable residual performance in the later cycles, although a significant fraction of the initial capacity is lost during the first few cycles, an aspect that should be considered for its long-term application⁶².

To evaluate the practical performance of PA@Pal₅₀₀, Cd(II) adsorption tests were conducted using simulated real water samples including potable water, wastewater, and seawater. Compared to the deionized water system, the RE % decreased with varying degrees in all three matrices (Table S4). Nevertheless, 1 mg/L Cd(II) concentration exhibited consistent RE % above 90%. Such a slight decrease was mainly attributed to the presence of minerals, organic matter, and trace salts in these water samples, which can interfere with Cd(II) adsorption by PA@Pal₅₀₀ as listed in Table S5⁶³.

To better understand how Cd(II) ions interact with PA@Pal₅₀₀ nanosorbent, a comparative FT-IR analysis was carried out before and after adsorptive capture. Initially, a broad absorption band at 3452 cm⁻¹ (Fig. 1a) was observed, corresponding to stretching modes of –OH and –COOH groups. These originate from residual structural –OH groups in Pal₅₀₀ and the –COOH groups in PA. The –OH and –COOH groups are negatively charged, while Cd(II) ions carry a positive charge, enabling electrostatic attraction between them⁶⁴. After adsorption, the FT-IR spectrum showed a noticeable shift in the –OH and –COOH stretching vibration peak from 3452 cm⁻¹ to 3471 cm⁻¹ (Fig. 1b). This shift indicates a change in the chemical environment of these functional groups, confirming electrostatic

interactions of Cd(II) with hydroxyl groups in Pal₅₀₀ as well as carboxylic groups in PA⁶⁵.

Beyond the electrostatic forces, PA contains oxygen- and nitrogen-bearing groups, with lone electron pairs capable of acting as ligands. These groups can coordinate bonds with Cd(II), to further enhance adsorption through complexation. The removal of Cd(II) occurs in two main stages: first, the ions migrate from the solution into the porous structure of PA@Pal₅₀₀ via successive mass transfer steps. Then, during the film transport stage, Cd(II) diffuses across the hydrodynamic boundary layer and penetrates the pore volume, where active sorbent interactions occur. Among these steps, pore diffusion is the rate-limiting process. Overall, the adsorption of Cd(II) on PA@Pal₅₀₀ follows a pore diffusion mechanism, consistent with the pseudo-first-order (PFO) kinetic model²⁹. Cd(II) ions can replace the exchangeable cations such as magnesium (Mg²⁺) on the surface of Pal₅₀₀ because they share similar charge and ionic size. This happens when Cd(II) ions in the surrounding solution interact with the Pal₅₀₀ surface and displace the cations that were previously adsorbed⁶⁶. The process is driven by four synergistic forces: (i) electrostatic attraction, (ii) coordination bonding, (iii) pore diffusion, and (iv) ion exchange, as illustrated in Scheme 1.

DFT and Monte Carlo simulation of Cd(II) ions removal by PA@Pal₅₀₀ nanosorbent

Density functional theory (DFT) calculations were performed using the DMol3 module in Materials Studio. After optimizing the geometries, the electrostatic potential, TDOS, and Fukui functions were examined to identify the key adsorption sites and clarify how Cd(II) ions interact with PA@Pal₅₀₀.

Electrostatic potential maps were used to examine how charge is distributed across Pal₅₀₀ and PA and to pinpoint the region most likely to bind Cd(II) ions. In both materials, the electrostatic potential was uneven, indicating the presence of distinct reactive sites that influence how Cd(II) interacts with each system. For the Cd(II)/PA system (Fig. 8a), it displayed localized negative potential regions, but these were concentrated primarily around particular donor atoms within the ligand⁶⁷. The optimized structure showed Cd(II) ions positioned closest to the nitrogen atom, indicating that nitrogen played a key role as a coordination center. This means that Cd(II) binding to PA is not governed by general surface interactions but instead by well-defined coordination sites, consistent with the chelating behavior of PA. In contrast, Cd(II)/Pal₅₀₀ complex (Fig. 8b), the maps showed well-defined positive and negative zones across the mineral surface, highlighting its heterogeneous electronic nature. The negatively charged regions, being electron-rich, served as the most favorable sites for Cd(II) attachment, driven by electrostatic attraction between the positively charged metal ions and these negatively polarized areas. This confirms that Cd(II) adsorption onto pal₅₀₀ occurs at specific, localized active sites rather than uniformly

across the surface⁶⁸. Overall, the electrostatic potential analysis suggests that Pal₅₀₀ offers a variety of heterogeneous surface sites for Cd(II) adsorption, whereas PA provides more specific, localized donor centers, especially around nitrogen for coordination. Thus, Cd(II) interaction with Pal₅₀₀ is best described as surface adsorption at active sites, while its interaction with PA follows a more defined coordination or chelation mechanism.

The TDOS results for both the Cd(II)/PA and Cd(II)/Pal₅₀₀ complexes show that Cd(II) binding favored noticeable modifications in their electronic structures. The appearance of electronic states near the Fermi level indicates that the interaction involves more than simple physical attraction; there are meaningful electronic contributions from the adsorption or coordination sites. The TDOS for the Cd(II)/PA complex (Fig. 9) reveals a more focused electronic response, pointing out to binding through specific donor atoms. This behavior aligns well with the known chelating properties of PA. In contrast, in the Cd(II)/Pal₅₀₀ system (Fig. 9), the TDOS pattern reflects a heterogeneous nanosorbent surface, where multiple active sites participate in capturing Cd(II) ions⁶⁹. Overall, the TDOS analysis supports two distinct binding modes: Pal₅₀₀ interacts with Cd(II) primarily through surface adsorption at varied active sites, while PA binds the metal ion through localized, well-defined coordination centers.

The Fukui function results (Table S6) show that in Model 1 (Cd/PA), the most reactive sites are concentrated around atoms O(20), O(22), O(25), and O(26). The nitrogen atoms N(8) and N(9) display noticeably lower reactivity, indicating that oxygen-bearing groups serve as the primary coordination centers for Cd(II) binding in PA. In Model 2 (Cd/Pal₅₀₀), however, the reactive regions are spread across multiple surface atoms, particularly Si(3), Si(24), O(70), and O(80). Such a distributed pattern suggests that Cd(II) adsorption on Pal₅₀₀ occurs at several heterogeneous active sites rather than being confined to a single coordination center. Overall, these findings support two distinct interaction modes: PA binds with Cd(II) through localized chelation at specific oxygen donors, while Pal₅₀₀ captures Cd(II) through a broader set of reactive surface sites⁷⁰.

Monte Carlo simulations were carried out using the Adsorption Locator annealing method, which produced five possible adsorption configurations for both Cd(II)/Pal₅₀₀ and Cd(II)/PA systems. In each case, the adsorption energies (E_{ads}) were negative, confirming that Cd(II) binding to both materials was thermodynamically favorable, as illustrated in Fig. 9. However, the adsorption energies obtained for the Cd(II)/Pal₅₀₀ complexes ranging from -3.959 to -3.540 kcal/mol were notably more negative than those calculated for the Cd(II)/PA system, which ranged from -1.492 to -1.220 kcal/mol⁷¹. This indicates that Pal₅₀₀ provides a more energetically stable environment for Cd(II) adsorption compared to PA. For both systems, the first and second configurations exhibited the most negative energies, meaning that they represent the most favorable binding modes. The small differences between these top-ranked energies suggest that more than one stable adsorption arrangement is feasible. Meanwhile, the less negative energies of the remaining configurations indicate lower, but still energetically favorable stability⁷².

The DFT and Monte Carlo sections have been substantially revised to connect them directly to the experimental adsorption trends and the proposed mechanism according to the following. Specifically: (i) the heterogeneous, spatially distributed electron-rich binding sites identified by the electrostatic-potential and Fukui analyses correspond to the energetically heterogeneous surface inferred experimentally from the Freundlich isotherm; (ii) the modest, negative Monte Carlo adsorption energies (physical/electrostatic range) are consistent with the predominantly physical adsorption indicated by the D–R mean free energy (<8 kJ/mol); (iii) the electrostatic nature of binding rationalizes the observed reduction in removal at elevated ionic strength; and (iv) the stronger Cd(II) binding to Pal₅₀₀ than to PA explains the dominant role of the clay surface in the composite. The computational interpretation was harmonized with the experimental mechanism, replacing the earlier

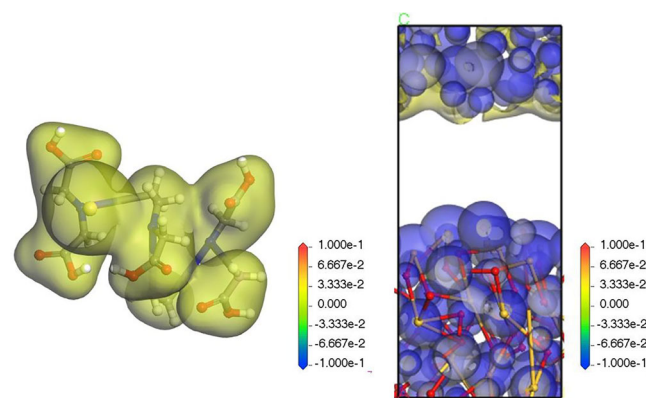
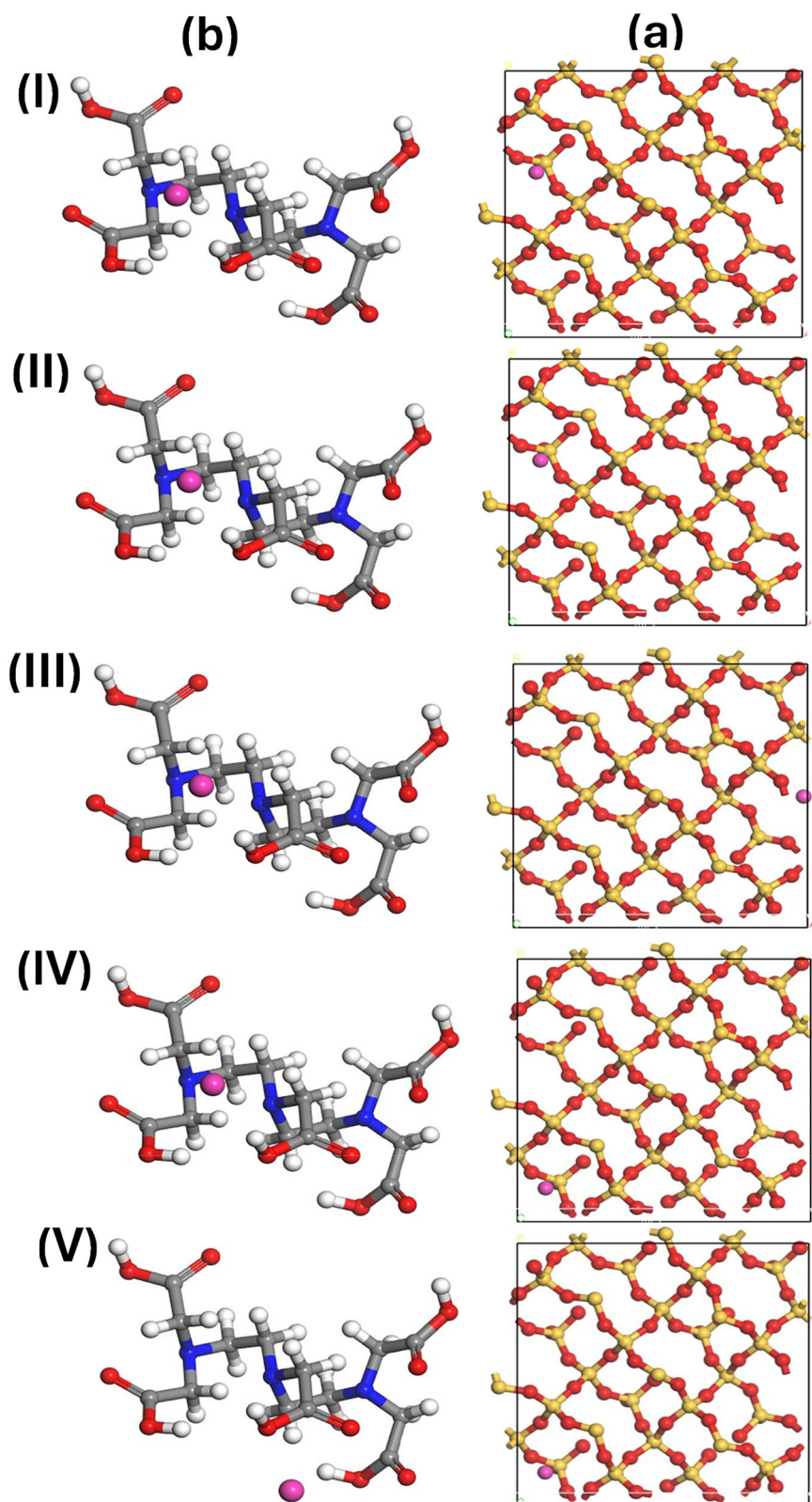


Fig. 8 | The ionic strength findings highlight the incorporated selectivity of PA@Pal₅₀₀ for Cd(II) ions, underscoring its practical applicability. For example, when the Na⁺ ion reached 25 mg/L, the RE% by PA@Pal₅₀₀ decreased by only 11% for 1 mg/L Cd(II) and 7% by 5 mg/L Cd(II) to refer to low contribution on removal effectiveness. Effect of ionic strength on adsorption Cd(II) onto PA@Pal₅₀₀ nanosorbent under numerous ionic strengths using solutions of NaCl (0 to 100 mg/L).

chemisorption/chelation wording with a consistent physical/electrostatic description supported by both datasets. Moreover, to link the computational results with the experimental adsorption behavior, the DFT and Monte Carlo findings are interpreted in the context of the measured isotherm, mean free energy, and ionic strength dependence. The electrostatic-potential and Fukui analyses show that Cd(II) binds to Pal₅₀₀ at multiple, spatially distributed electron-rich (negatively polarized) sites, principally oxygen and silicon surface centers, rather than at a single uniform site. This directly mirrors the experimental finding that the equilibrium data follow the Freundlich rather than the Langmuir model, i.e., adsorption on an energetically heterogeneous surface bearing a distribution of non-equivalent sites. The Monte Carlo adsorption energies are modest and negative (-14.8 to -16.6 kJ/mol for Cd/Pal₅₀₀ and -5.1 to -6.2 kJ/mol for Cd/PA), consistent with physical/electrostatic interactions and in agreement with the experimentally derived D–R mean free energy (<8 kJ/mol), which likewise indicates that Cd(II) uptake is governed predominantly by physical (electrostatic) adsorption rather than chemisorption. The electrostatic nature of the dominant Cd(II)/Pal₅₀₀ interaction further rationalizes the measured decline in removal with increasing ionic strength, as competing Na⁺ ions screen and partially occupy the same negatively charged sites. Finally, the more negative adsorption energies for Cd/Pal₅₀₀ than for Cd/PA indicate that, within the PA@Pal₅₀₀ composite, the palygorskite surface provides the dominant electrostatic binding sites while the PA component contributes additional localized oxygen-donor sites; together these account for the multi-site, heterogeneous uptake inferred from the isotherm.

In summary, the PA@Pal₅₀₀ nanosorbent effectively removed Cd(II) from water under optimal conditions (pH 6, 50 mg dose). The equilibrium and kinetic data were best described by the Freundlich isotherm and the pseudo-first-order model (both $R^2 > 0.98$), indicating rapid uptake on an energetically heterogeneous surface, while the Dubinin–Radushkevich mean free energy (<8 kJ/mol) identified a predominantly physical/electrostatic mechanism; the process was found to be spontaneous and endothermic. Increasing ionic strength (Na⁺) lowered but did not suppress Cd(II) removal, confirming selectivity for Cd²⁺. The adsorbent was reusable over repeated cycles, retaining about 42% of its initial efficiency after nine adsorption–desorption cycles with a stable performance beyond the fifth cycle, and it achieved 95%, 93%, and 92% Cd(II) removal from tap water, wastewater, and seawater, respectively. The electrostatic potential, density-of-states, Fukui, and Monte Carlo analyses indicated that Cd(II) binds mainly at the multiple electron-rich sites of Pal₅₀₀, with the PA component contributing additional localized oxygen-donor sites. Overall, PA@Pal₅₀₀ is a promising, reusable

Fig. 9 | The electrostatic potential maps identify electron-rich (negative potential) and electron-deficient (positive potential) regions, highlighting the preferred coordination sites for Cd(II). These results provide insight into the electronic interactions governing Cd(II) adsorption on the PA-functionalized silico-oxygen surface. Molecular structures and electrostatic potential (ESP) distributions of the **a** Cd(II)/PA system and **b** Cd(II)/silico-oxygen tetrahedron system.



nanosorbent for trace Cd(II) removal from real water matrices as compared to other previously reported and studied adsorbents as listed in Table 3^{73–75}.

Methods

All employed chemicals and reagents are selected from analytical reagent grades, as depicted in Table S1.

Instrumentations

The utilized techniques for characterization of the assembled PA@Pal₅₀₀ nanosorbent are outlined in Table S2.

Preparation of PA@Pal₅₀₀ sorbent

Palygorskite clay (Pal) was thermally modified at 500 °C (Pal₅₀₀) as previously [29]. In sequential steps, 3.0 g of PA was added to 7.0 g of (Pal₅₀₀)

Table 3 | Superiority of PA@Pal500 vs other previously reported and studied adsorbents in removal of Cd(II) ions

Adsorbent	Modification method	Studied parameters	Removal performance	Regeneration	Theoretical evaluation	Application on real samples	Ref
Mercapto-palygorskite functional materials	MPTMS grafting (–SH) for 30 min	pH: 2.0–8.0 Cd(II): 0–50 mg/L Time: 20 min Interf ions: K ⁺ , Ca ²⁺	Cap = 30.69 mg/g	Three regeneration cycles	DFT study	None	25
Mercapto- and amino-functionalized palygorskite	Nanotexturization in aqueous gel	pH: 6.6 Time: 60 min Mass: 0.025 g Cd(II): 0–100 mg L ^{−1} Time: 0–120 min Temp: 15.0–35.0 °C	Cap = 27.46 mg/g	None	None	None	26
MgO-modified palygorskite/rice straw biochar	Impregnation reaction for >5 h	pH: 2.0–7.0 Time: 11–720 min Mass: 1.0 to 3.0 g/L	Cap = 441.1 mg/g	None	DFT computation	None	70
MgO/Attapulgit	Impregnation-aging for 13 h	pH: 2–6 Time: 1–300 min Cd(II): 1–50 mg/L Temp: 288–308 K	Cap = 121.14 mg/g	None	None	None	73
MgO-biochar derived from agricultural residue	Impregnation-pyrolysis process in nitrogen flow for 2 h	pH: 3–6 Cd(II): 50–100 mg/L Time: 24 h Temp: 25 °C Dosage: (0.5 g/L)	R% = 97.1%	Five regeneration cycles	None	None	74
Natural brine MgO/palygorskite	Preparation for 3 h	pH, 5.50 Dosage: 1 g/L Cd(II): 200–1000 mg/L	R% = 98.69%	One regeneration cycle	None	None	75
PA@Pal500	Microwave synthesis for 15 min	pH: 2–7 Cd(II): 1–100 mg/L Time: 2–120 min Dosage: 10–100 mg Temp: 288–328 K Ionic strength: 10–100 mg NaCl	R% ≥ 97%	Nine regeneration cycles	DFT and Monte Carlo simulations	89–100% removal efficiency from tap, sea and wastewater	This work

**Scheme 2 | Preparation of PA@Pal₅₀₀ nanosorbent.**

and blended in 100 mL beaker, and 10 mL double distillate water (DDW) was added and well stirred. Then the blend was irradiated in a microwave oven for 15 min to produce PA@Pal₅₀₀ nanosorbent near water dryness. The washing and heating procedures were repeated three times to remove excess acids, and the produced PA@Pal₅₀₀ nanosorbent was then dried at 80 °C for 6.0 h. The schematic diagram for synthesis of PA@Pal₅₀₀ nanosorbent is illustrated in Scheme 2.

Batch adsorption experiments for removal of Cd(II) ions

The adsorptive capture performance of Cd(II) by PA@Pal₅₀₀ was conducted via the batch method at 25.0 °C using a shaker at 300 rpm and 30 min. Typically, a specified mass of PA@Pal₅₀₀ was incorporated into Cd(II) ions (1.0 and 5.0 mg/L solutions) with a defined volume (50 mL) in triplicate determinations. The residual Cd(II) ions were analyzed by the dithizone method at $\lambda_{\max} = 515$ nm. The Cd(II) ions removal efficiency (RE %) and quantity of (Q_e mg/g) at equilibrium were identified from Eq. (2) and Eq. (3), respectively (Table 4). The values of RE % of Cd(II) ions were investigated by some significant factors including diverse pH levels, time, Cd(II) concentration, PA@Pal₅₀₀ dosage, NaCl electrolytes, and temperature of reaction.

The 10–100 mg PA@Pal₅₀₀ were investigated to characterize the impact of dosage. The dosage was thoroughly reacted with 50 mL of Cd(II) solutions (1.0 and 5.0 mg/L) for 30 min.

A 0.1 M HNO₃ or NaOH solution was used to control the pH of the Cd(II) solution from 1 to 7. A 20 mg of PA@Pal₅₀₀ was mixed with 50 mL of Cd(II) ions (1.0 and 5.0 mg/L) for 30 min.

The kinetics of Cd(II) (1.0 and 5.0 mg/L) interaction with PA@Pal₅₀₀ were accomplished at 5–60 min using 20 mg of PA@Pal₅₀₀ with 50 mL of Cd(II) (1.0 and 5.0 mg/L), and pH 6.

A 50 mL of 1 to 10 mg/L Cd(II) ions was shaken with 20 and 50 mg of PA@Pal₅₀₀ for 30 min at pH 6. The assessment and evaluation of the mechanism was also established PA@Pal₅₀₀ via selected adsorption isotherm models.

Table 4 | Non-linear kinetic models and parameters for adsorption of Cd(II) ions with standard deviations within ± 0.5 – 3.1%

Kinetic model	Pseudo-first order				Pseudo-second order			Intraparticle diffusion			Elvoich		
Plot	Q_t vs time (t)				Q_t vs time (t)			Q_t vs time (t)			Q_t vs time (t)		
Constant	q_e (exp)	K_1	q_e (calc)	R^2	K_2	q_e (calc)	R^2	K_{diff}	C	R^2	β	α	R^2
Units	mg g ⁻¹	min ⁻¹	mg g ⁻¹	-	mg g ⁻¹ min ⁻¹	mg g ⁻¹	-	mg g ⁻¹ min ^{-1/2}	mg g ⁻¹	-	-	mg g ⁻¹ min ⁻¹	-
1 mg/L	2.42	0.039	2.49	0.991	0.014	3.08	0.993	0.25	0.107	0.933	1.24	0.179	0.975
5 mg/L ¹	10.75	0.025	11.57	0.982	0.0014	15.48	0.983	1.07	9.99E-27	0.972	0.217	0.420	0.983

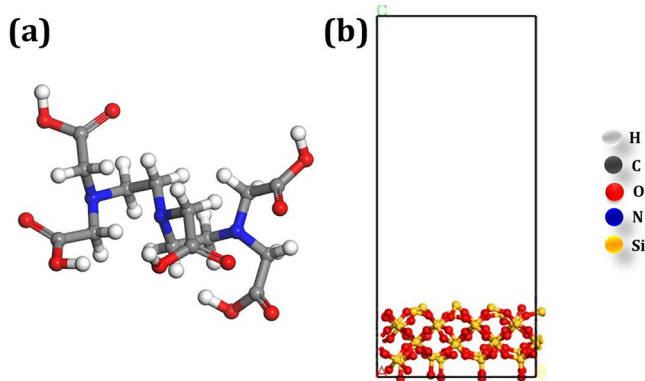


Fig. 10 | The configurations represent the most energetically favorable adsorption geometries identified during the simulation, illustrating the preferred interaction modes of Cd(II) with the Pal₅₀₀ surface and the PA molecule. These optimized adsorption structures were used to analyze the adsorption mechanism and identify the most stable binding sites. Five low-energy adsorption configurations of the **a** Cd(II)/Pal₅₀₀ system and **b** Cd(II)/PA system obtained by Monte Carlo simulated annealing using the Adsorption Locator module.

The range 15–55 °C was optimized by mixing 20 mg of PA@Pal₅₀₀ with 50 mL of Cd(II) ions solution (1.0 and 5.0 mg/L) for 30 min at pH 6. The thermodynamic parameters for Cd(II) uptake onto PA@Pal₅₀₀ were also computed from the gathered data.

The ionic strength impact on Cd(II) ions elimination by PA@Pal₅₀₀ was investigated in the presence of mixed 10–100 mg/L NaCl with 20 mg of PA@Pal₅₀₀ and 50 mL of Cd(II) ions (1.0 and 5.0 mg/L) for 30 min at pH 6.

The reusability of the PA@Pal₅₀₀ was conducted by reacting 100 mg PA@Pal₅₀₀ with Cd(II) ions (50 mL of 10 mg/L) at 30 min and pH 6. The loaded Cd(II) on PA@Pal₅₀₀ was centrifuged and rinsed with 0.01 M HCl, ethyl alcohol, and finally with DDW, after which the leftover material was dried for further application.

The efficacy of PA@Pal₅₀₀ in the extraction of Cd(II) ions from wastewaters was also explored in this study. 50 mg of PA@Pal₅₀₀ was agitated with 50 mL Cd(II) ions (1.0 and 5.0 mg/L) spiked into tap water, seawater and wastewater for 30 min.

Geometry optimization, electrostatic potential, adsorption energy, density of states, and Fukui function calculations were conducted using the DMol³ package. The palygorskite model was constructed from a silico-oxygen tetrahedral framework derived from the experimental XRD structural parameters, so that the simulated surface reproduces the pyrolyzed Pal₅₀₀ material; it was expanded to a (2 × 2 × 1) supercell and separated by a 27 Å vacuum layer along the z-direction to eliminate spurious interlayer/periodic-image interactions and to expose the surface oxygen and Si–O sites involved in Cd(II) binding. The DFT calculations (geometry optimization, electrostatic potential, density of states, Fukui indices, and adsorption energy) were carried out in DMol³ within the GGA-PBE framework, using the DNP 4.4 basis set for valence electrons and the DSPP treatment for core electrons, with a SCF tolerance of 1×10^{-5} eV (maximum 1000 iterations) a level of theory widely applied to clay and oxide surfaces and adequate for

the qualitative site-screening intended here. The adsorption energy (E_{ads}) was calculated according to Eq. (16)³⁰. The Cd(II) adsorption behavior on PA@Pal₅₀₀ was then investigated by Monte Carlo simulated annealing (Adsorption Locator, Materials Studio 2020). The PA molecule and the Pal₅₀₀ supercell were first fully optimized with the Forcite module (smart algorithm; convergence thresholds of 2×10^{-5} kcal/mol for energy, 0.001 kcal/mol/Å for force, 0.001 GPa for stress, and 1×10^{-5} Å for displacement). The relaxed structures of PA and Pal₅₀₀ super-cell are presented in Fig. 10a, b. Because the calculations are gas-phase and do not include explicit solvation or surface hydroxylation, they are used to identify preferred binding sites and qualitative trends in support of the experimental mechanism, not as absolute aqueous adsorption energies.

Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Received: 4 May 2026; Accepted: 25 July 2026;

Published online: 22 September 2026

References

- Elawady, A., Mohamed, J. A. H. & Abid, M. B. Advancements in microplastics detection techniques and their multidimensional impacts on aquatic ecosystems and human health. *J. Hazard. Mater. Plast.* **2**, 100025 (2026).
- Guan, Y. et al. High-efficiency removal of Cu²⁺, Zn²⁺, and Mg²⁺ from contaminated water using a conductive hydrogel based on gellan gum/Ti₃C₂ MXene/PEDOT-S:H/aluminum oxide silicate composite. *Chem. Eng. Sci.* **332**, 124091 (2026).
- Li, X. et al. Coffee shell as a green reductant for iron recovery from pyrite cinder: synergy of biomass-driven reduction and acid leaching. *Min. Eng.* **234**, 109747 (2025).
- Saxena, V. Water quality, air pollution, and climate change: investigating the environmental impacts of industrialization and urbanization. *Water Air Soil Pollut.* **236**, 73 (2025).
- Shaukat, A. et al. Arginine-derived green-synthesized carbon dot@copper sulfide nanorods (CD@CuS) as a potent and biocompatible antibacterial agent: experimental and computational insights. *Inorg. Nano-Met. Chem.* **56**, 526–537 (2026).
- Qezelje, H. H. et al. Synthesis of a Chlorella/ZnO/ZnFe₂O₄ green nanocomposite for preconcentration of heavy metals in food samples via ICP-OES detection. *Food Chem. X* **36**, 103852 (2026).
- Ahmad, K. et al. Arsenic contaminants of groundwater and its mitigation. In *Soil, Water Pollution and Mitigation Strategies: A Spatial Approach* 389–415 (Springer, 2024).
- Shetty, S. S. et al. Environmental pollutants and their effects on human health. *Heliyon* **9**, e19496 (2023).
- Dehkordi, M. M., Nodeh, Z. P., Dehkordi, K. S., Salmanvandi, H., Khorjastan, R. R. & Ghaffarzadeh, M. Soil, air, and water pollution from mining and industrial activities: sources of pollution, environmental impacts, and prevention and control methods. *Results Eng.* **23**, 102729 (2024).

10. Oladimejia, T. E. et al. Review on the impact of heavy metals from industrial wastewater effluent and removal technologies. *Heliyon* **10**, e40370 (2024).
11. Jomova, K., Alomar, S. Y., Nepovimova, E., Kuca, K. & Valko, M. Heavy metals: toxicity and human health effects. *Arch. Toxicol.* **99**, 153–209 (2025).
12. Wang, C., Chen, Y., Liu, Y., Ma, B. & Wang, C. From hazardous waste to resource: proper disposal of cadmium. *Sep. Purif. Technol.* **377**, 134304 (2025).
13. Wang, D. et al. Mechanistic insights into hexamethylphosphoramide degradation and phosphorus recovery via mZVI/O₂: beyond conventional oxidation through iron-phosphorus speciation. *Chem. Eng. J.* **538**, 176658 (2026).
14. Liang, S. et al. Catalysis of nonstoichiometric Mn–Co spinel oxides in effective peroxymonosulfate activation: structure–activity relationship and flow-through organic wastewater treatment. *Sep. Purif. Technol.* **397**, 138142 (2026).
15. Rasin, P. et al. Exposure to cadmium and its impacts on human health: a short review. *J. Hazard. Mater. Adv.* **17**, 100608 (2025).
16. Robert, B. & Kalpathy, S. K. Future prospects, challenges and opportunities in detecting heavy metal ions in water. *J. Water Process. Eng.* **75**, 107992 (2025).
17. Wazeer, I. et al. Comparative analysis of lead and cadmium extraction capacities of hydrophobic deep eutectic solvents. *J. Ind. Eng. Chem.* **143**, 303–318 (2025).
18. Zhou, L. Improving the removal performance of cadmium from wastewater by phosphate-modified sludge biochar: a mineral dissolution-precipitation perspective. *Desal. Water Treat.* **320**, 100865 (2024).
19. Bhattacharyya, K., Sen, D., Banik, A. K. & Ganguly, S. Adsorptive removal of cadmium from aqueous medium—a critical review. *Phys. Chem. Earth (Pt. A/B/C)* **134**, 103538 (2024).
20. Achchar, K., Valange, S., Kherbeche, A., Gallard, H. & Draoui, K. Cadmium removal using Rhassoul/alginate composite beads: isotherms, kinetics, and thermodynamic study. *Adsorption* **31**, 1–21 (2025).
21. Muleta, W. S. Enhanced cadmium removal from aqueous solution via electro coagulation using calcium oxide nanoparticles based natural coagulant: kinetics study and RSM optimization. *S. Afr. J. Chem. Eng.* **54**, 99–113 (2025).
22. Zare, S. & Pouranfard, A. Membrane nanofiltration of cadmium-containing wastewater by a modified polysulfone membrane: experimental and statistical analysis. *Int. J. Environ. Sci. Technol.* **22**, 16585–16604 (2025).
23. Arizpe-Díaz, D. O. et al. Evaluation of cadmium removal in an aqueous solution by biosorption in a batch system with banana, peanut, and orange husks. *Environments* **12**, 97 (2025).
24. Hamid, N. H. A. et al. A state-of-art review on the sustainable technologies for cadmium removal from wastewater. *Water Reuse* **14**, 312–341 (2024).
25. Gao, G., Sun, T., Sun, Y., Xu, Y. & Liang, X. Adsorption of Cd(II) on mercapto-palygorskite functional materials with different proportions: Performance, mechanism and DFT study. *J. Clean. Prod.* **452**, 142052 (2024).
26. Liang, X. et al. Sorption of Cd²⁺ on mercapto and amino functionalized palygorskite. *Appl. Surf. Sci.* **322**, 194–201 (2014).
27. Abara, H. et al. Eco-friendly cadmium removal using novel modified clay/alginate floatable beads: a sustainable solution for water pollution mitigation. *Arab. J. Sci. Eng.* **50**, 6537–6552 (2025).
28. Mahmoud, M. E., Obada, M. K. & Nabil, G. M. Enhanced removing of phosphate ions from agricultural drainage wastewater by using microwave-assisted synthesized attapulgite (Fullers earth)@ carboxymethylcellulose nanocomposite. *Int. J. Biol. Macromol.* **255**, 128081 (2024).
29. Nabil, G. M., Obada, M. K. & Mahmoud, M. E. Implementation of integrated Palygorskite with copper–aluminum layered double hydroxides composite in superior nitrate pollutant removal from wastewater. *J. Ind. Eng. Chem.* **151**, 605–617 (2025).
30. Li, X., Chen, W., Jiang, J., Zhu, X. & Zhao, P. Preparation of composite material sodium dodecyl sulfonate/2-methylimidazole zinc MOF/palygorskite and its high-efficiency adsorption for tetracycline. *Appl. Clay Sci.* **279**, 108057 (2026).
31. Wang, J. et al. A green approach for fabricating palygorskite-based aerogel with enhanced adsorption performance for dye removal. *Sep. Purif. Technol.* **373**, 133429 (2025).
32. Xiang, X., Li, X., Yang, T., Cao, J. & Yang, Z. A novel Fe–Ca composite modified biochar prepared for enhanced Cd adsorption: experimental, DFT calculation and mechanism unveiling. *J. Water Process Eng.* **68**, 106565 (2024).
33. Xu, C., Zhu, C., Li, Y., Zhang, H. & Lv, W. Palygorskite-mediated simultaneous nutrient removal and antibiotic degradation from aquaculture wastewater in lab-scale constructed wetlands. *Chem. Eng. J.* **499**, 156568 (2024).
34. Rusmin, R. et al. Facile one pot preparation of magnetic chitosan-palygorskite nanocomposite for efficient removal of lead from water. *J. Colloid Interf. Sci.* **608**, 575–587 (2022).
35. Zhang, S., Zhong, L., Yang, H., Tang, A. & Zuo, X. Magnetic carbon-coated palygorskite loaded with cobalt nanoparticles for Congo Red removal from waters. *Appl. Clay Sci.* **198**, 105856 (2020).
36. Hou, L. et al. Pentetic acid/ammonia cooperatively stabilizes Cu(II) as an efficient oxidant for green thiosulfate leaching of gold. *Miner. Eng.* **218**, 109043 (2024).
37. Liu, Y. et al. Multiple removal mechanism of EDTA-enteromorpha-chitosan-based hydrogel for organic/inorganic cation-anion complex pollution. *Int. J. Biol. Macromol.* **339**, 149953 (2025).
38. Fani, K. et al. Upgrading residual fatty biomass to renewable diesel over Ni based catalysts supported on natural palygorskite: the effect of promoters and reaction conditions. *Biomass Bioenergy* **207**, 108703 (2026).
39. Zhang, S., Cao, M., Luo, J., Li, Y. & Tian, H. Ga₂O₃-ZnO/palygorskite-based SAPO₄⁻³ molecular sieves for catalytic hydrogenation of CO₂ to light olefins. *Colloids Surf. A Physicochem. Eng. Asp.* **732**, 139208 (2026).
40. Alves, J. L. F. et al. Screening of alternative catalysts for upgrading pyrolysis vapors from *Chlorella* sp. to hydrocarbon-rich bio-oil as a strategy for valuing naturally occurring materials and industrial wastes. *Renew. Energy* **256**, 124620 (2026).
41. Wang, C.-X., Wang, W., Wang, N. & Zhang, X.-F. Durable and multifunctional superhydrophobic composite coating based on activated PAL for active protection of aluminum alloy. *Appl. Surf. Sci.* **716**, 164717 (2026).
42. Liu, L. et al. The cadmium removal potential of crab shell biochar: performance and quantitative mechanisms analysis. *Colloids Surf. A Physicochem. Eng. Asp.* **730**, 139064 (2026).
43. Tang, C. et al. Adsorption mechanism of simultaneous removal of Pb(II), Cd(II) and As(III) by magnetite-modified attapulgite. *Colloids Surf. A Physicochem. Eng. Asp.* **731**, 139114 (2026).
44. Faizan, M. et al. Mesoporous magnetic MnFe₂O₄@SiO₂-chitosan nanocomposite for efficient adsorptive removal of Zn(II) and Cd(II) ions from aqueous media. *Desal. Water Treat.* **324**, 101488 (2025).
45. He, J., Gun, M., Wang, Y., Mo, S. & Long, F. Synthesis of CaCO₃ microspheres modified by glycine and ethylene glycol for adsorption removal of Cd(II). *Environ. Res.* **285**, 122442 (2025).
46. Ji, H. et al. Simultaneous removal of Cr(VI), Cd(II) and As(III) in groundwater by ZVI-biochar based composite: synergy, performance and mechanism. *Process Saf. Environ. Prot.* **201**, 107460 (2025).
47. Shewatatek, S., Gonfa, G., Hailegiorgis, S. M. & Tessema, B. Adsorptive removal of lead ions from wastewater using modified diatomite. *J. Hazard. Mater. Adv.* **20**, 100900 (2025).
48. Jaffar, F. H. et al. Innovative chicken eggshells-derived bio-hydroxyapatite/mullite ceramic hollow fiber adsorptive membrane for efficient cadmium removal. *Ceram. Int.* **51**, 43947–43967 (2025).

49. Huang, H. et al. Maghemite-biochar/hypochlorite synergy for enhanced thallium, lead, and cadmium removal. *Desal. Water Treat.* **324**, 101507 (2025).
50. Saaïd, A. A. & Ibrahim, B. M. Dibenzo-18-crown-6 modified lignin as a sustainable adsorbent for removal of Lead and cadmium ions from aqueous solutions. *Int. J. Biol. Macromol.* **309**, 142906 (2025).
51. Hajjar, D., Mohery, M. & Mahran, G. M. A. Ecofriendly method for removal of cadmium ions and synthesis of cadmium oxide nanoparticles using a new generation of amine-functionalized cellulose. *RSC Adv.* **15**, 38470–38493 (2025).
52. Fan, J. et al. Synthesis of magnetic zeolite 4A using thermally reduced red mud and metakaolin via ultrafast Joule heating for heavy metal ion removal from industrial wastewater. *Micropor. Mesopor. Mater.* **403**, 114019 (2026).
53. Zhang, R. et al. Analysis of the adsorption behavior and mechanism of acylhydrazine functionalized β -cyclodextrin: efficient removal of harmful dyes, heavy metal ions, and sulfides from wastewater. *Colloids Surf. A Physicochem. Eng. Asp.* **734**, 139431 (2026).
54. Baba, I. A., Amawuosa, E. C., Mustapha, S., Abdulkareem, A. S. & Tijani, J. O. Adsorption kinetics, isotherm, and thermodynamics studies of phytochemical-assisted synthesized ZnO nanoparticles for the removal of selected heavy metals from abattoir wastewater. *Desal. Water Treat.* **325**, 101628 (2026).
55. Mudigonda, S. et al. Simultaneous removal of multiple heavy metal ions from simulated wastewater using Nanoceria-Carbon Nanotubes infused three-dimensional novel cryogels. *J. Water Process. Eng.* **81**, 109376 (2026).
56. Shen, S. et al. Mechanistic insights into heavy metal adsorption on hydroxyapatite: new perspectives from dynamic adsorption behavior. *J. Environ. Chem. Eng.* **13**, 118855 (2025).
57. Huang, X. et al. Preparation of KMnO_4 modified N-doped bamboo biochar by microwave pyrolysis for enhancing heavy metals adsorption from aqueous solution. *J. Contam. Hydrol.* **276**, 104755 (2026).
58. Ajala, E. O. et al. Adsorption of lead and chromium ions from electroplating wastewater using plantain stalk modified by amorphous alumina developed from waste cans. *Sci. Rep.* **14**, 6055 (2024).
59. Wang, J., Zheng, Z., Liu, L. & Pan, J. Self-diffusing halloysite-based adsorbent for efficient and selective lead removal. *Sep. Purif. Technol.* **382**, 135651 (2026).
60. Shamseddine, H., Aad, N. A., Younes, G. & Al-Oweini, R. Europium-doped silica nanoparticles for efficient and sustainable lead removal from aqueous solutions. *Result Mater.* **28**, 100826 (2025).
61. Wang, W., Zhang, G., Dong, L. & Gu, P. EDTA-modified hydroxyl-functionalized covalent organic frameworks for enhanced Pb(II) removal from complex wastewater: adsorption mechanism and performance. *Sep. Purif. Technol.* **389**, 136818 (2026).
62. Almousa, M. et al. Development of Fe–Zr hybrid MOFs for high-efficiency arsenic removal in aqueous systems. *Results Eng.* **29**, 108858 (2026).
63. Luo, H., Li, K., Yang, X. & Xiang, H. MgAlFe-LDHs-based self-healing composite hydrogel for efficient removal of multiple dyes and density functional theory calculations. *Sep. Purif. Technol.* **384**, 136318 (2026).
64. Liu, B., Han, X., Yang, K., Zhang, J. & Cai, H. Lignin-intercalated WS_2 with synergistic adsorption for efficiency lead removal. *Bioresour. Technol.* **439**, 133284 (2026).
65. Li, S. et al. In-situ preparation of lignin/ Fe_3O_4 magnetic spheres as bifunctional material for the efficient removal of metal ions and methylene blue. *Int. J. Biol. Macromol.* **259**, 128971 (2024).
66. Liu, L. et al. In situ integrated removal of tetracycline and cadmium from soil/water by the voltaic-cell-effect of the iron (II) sulfide-hydrogel. *Appl. Surf. Sci.* **715**, 164589 (2026).
67. Babeker, T. M. A., Lv, S., Khalil, M. N. & Hao, Z. Biochar modified by ammonium pyrrolidine dithiocarbamate for high selective adsorption of copper in wastewater. *Sep. Purif. Technol.* **354**, 129436 (2025).
68. Nabat, K. Y. et al. A dual-function Schiff base ligand: fluorescent probe and high sustainable adsorption via PVC immobilized for Cd(II) and Pb(II): experiments, DFT calculations and cost analysis. *Chem. Eng. J.* **516**, 164211 (2025).
69. Yan, K. et al. Unraveling immobilization mechanisms of Cd in soil by MgO-modified palygorskite/biochar composite: DFT calculation and combined-artificial aging. *J. Environ. Manage.* **370**, 122576 (2024).
70. Yang, K. et al. Adsorption performance and mechanisms of MgO-modified palygorskite/biochar composite for aqueous Cd (II): experiments and theoretical calculation. *Appl. Surf. Sci.* **638**, 157965 (2023).
71. Zhang, L. et al. Integrating spectroscopic analysis and theoretical calculations to elucidate the adsorption efficiency and mechanisms of Cd, Pb, and Cu using novel carboxymethyl cellulose/pectin-based hydrogel beads. *Int. J. Biol. Macromol.* **305**, 141028 (2025).
72. Zheng, B., He, Y., Tian, S., Wang, S. & Tang, L. Bamboo fiber-reinforced chitosan biomass aerogel for Cr(VI) and Cu(II) capture: batch adsorption experiments and density functional theory calculations. *Appl. Surf. Sci.* **703**, 163391 (2025).
73. Wang, H., Wang, X., Ma, J., Xia, P. & Zhao, J. Removal of cadmium (II) from aqueous solution: a comparative study of raw attapulgite clay and a reusable waste–struvite/attapulgite obtained from nutrient-rich wastewater. *J. Hazard. Mater.* **329**, 66–76 (2017).
74. Liang, H. et al. Facile fabrication of highly porous MgO-modified biochar derived from agricultural residue for efficient Cd(II) removal from wastewater. *Inorg. Chem. Commun.* **154**, 110900 (2023).
75. Tian, G., Wang, W., Zong, L. & Wang, A. MgO/palygorskite adsorbent derived from natural Mg-rich brine and palygorskite for high-efficient removal of Cd(II) and Zn(II) ions. *J. Environ. Chem. Eng.* **5**, 1027–1036 (2017).

Acknowledgements

The authors would like to acknowledge the support of Alexandria University.

Author contributions

Mohamed E. Mahmoud, Gehan M. Nabil, Mohammed K. Obada, and Magda E. Abouelanwar have equally contributed to this work.

Funding

Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).

Competing interests

The authors declare no competing interests.

Consent to publish

The authors of this manuscript agree to submit this manuscript for publication in *npj Clean Water*.

Consent to participate

Not applicable.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41545-026-00616-0>.

Correspondence and requests for materials should be addressed to Mohamed E. Mahmoud.

Reprints and permissions information is available at
<http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026